C3: Committed Confidence Calibration

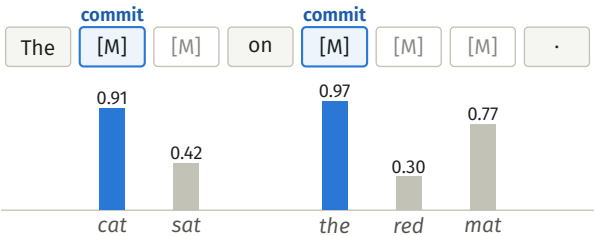
Selection bias and two sources of error in confidence-based token commitment in diffusion language models

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Diffusion LMs write by filling in blanks, surest first

Each step the model guesses every blank at once and keeps the k guesses it is surest of



bar: confidence c_i of the best guess per blank; the top $k = 2$ are kept

- The model starts from all-blank text and fills it in over a few steps.
- Top-k (LLaDA-8B, Dream-7B) keeps the k surest guesses; Fast-dLLM keeps all guesses above a threshold.
- A kept token won a contest of noisy scores: the setting of the winner's curse.

Question. When the model is 90% sure of a token it chose, is that token right 90% of the time? And when it fills several blanks in one step, which kind of error dominates?

Filling several blanks in one step can go wrong in two ways

Total error of a set S of tokens committed together, given the visible text

$$\mathrm{KL}\Big(p^* \,\Big\|\, \prod_{i\in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\mathrm{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}\big(p_i^* \,\|\, p_{\theta,i}\big)}_{\text{(B) estimation}}$$

- **(A)** Dependency: blanks are guessed separately, so the guesses can clash, e.g. *New* + *Angeles*.
- **(B)** Estimation: each blank's own probabilities are wrong. Work on dependency in parallel decoding treats **(B)** as fixed.
- C2 measures both where p^* is known; C1 asks whether chosen tokens are overconfident.

Work	Checks confidence of chosen tokens?	Separates (A) and (B)?
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	positional only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Tang et al. (2026)	no	identifies
This work	**yes**	**both**

Three contributions

C1

Are chosen tokens overconfident?

Is a token that *won* against other blanks right less often than its confidence says? Tested on LLaDA-8B, Dream-7B (Base and Instruct) and our own 110M model.

C2

How large are (A) and (B)?

Measured exactly on a synthetic JSON task where the true answer distribution is known. On real text, evaluated using disagreement and calibration proxies.

C3

A corrected commit rule

Correct the model's confidence, then commit every token that clears a bar. No extra model calls. Compared with six decoders at equal speed on GSM8K (math) and MBPP (code).

Theory: when can winning make a token overconfident?

s: a text with blanks M; ℓ_i: score of the guess at blank i; $a_i = 1$ if it is right

Proposition 1 · Winning alone adds no bias

If each blank's score and correctness are independent of the other blanks' given *s*, and the contest P is drawn independently of them, the winner i of P satisfies

$$\Pr\left(a_i = 1 \mid \ell_i, s, i = \arg\max_{j \in P} \ell_j\right) = \pi(\ell_i, s).$$

A winner is right as often as any token with its score and context. So unequal noise across blanks cannot by itself bias winners

Proposition 2 · Winners' overconfidence has two parts

$\Delta(m)$: confidence minus accuracy of tokens that won among m blanks.

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\text{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\text{field term}}$$

$\pi(\ell)$: usual calibration curve; $\pi(\ell, s)$: the same curve given the context. The field term is nonzero only if accuracy at a given score depends on the context; with $m = 1$ (no contest) it is exactly 0.

So a **winner's curse** can come only from the field term

C1

Gap = confidence − accuracy)

E1 · gap of winners

$\Delta(m)$: average gap of tokens that won among m blanks ($m \le 32$). E1 = $\Delta(32) - \Delta(1)$.

E2 · field term

The winner's-curse part of the gap (Proposition 2). E2 = field(32)−field(1).

E3 · what decoders leave

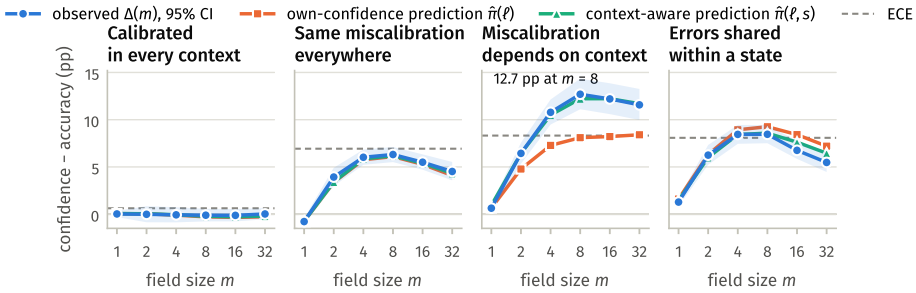
Decoders fill easy blanks first. At equal confidence, are the blanks they leave right less often than random ones?

Hypotheses. **H1** E1 > 0; **H2** E2 > 0; **H3** blanks a decoder leaves are more overconfident than random ones.

A cross-check by regression: on winning tokens, predict correctness from the score ℓ_i and the rank in the contest, with and without one offset α_s per text. A winner's curse shows as a negative rank effect β_z without the offsets that disappears with them.

Test on simulated data (T0)

Four simulated worlds with known answers



Why ECE is not enough. When overconfidence depends on context, ECE (the usual calibration score) remains near zero, whereas E1 clearly captures the overconfidence.

C1 results: does winning make a token overconfident? (E1, E2)

Seven model/task pairs; blanks chosen at random (uniform) or left by a top-k decoder (k=4)

	E1 = Δ(32) − Δ(1)		E2 = field contrast	
Model	uniform	decoder	uniform	decoder
LLaDA Base	−0.003	0.006	0.0010	0.0036
Dream Base	−0.002	0.015	0.0011	0.0029
LLaDA-I GSM8K	−0.022	0.024	−0.003	0.021
LLaDA-I MBPP	0.025	0.156	0.013	0.036
Dream-I GSM8K	0.061	0.072	0.021	0.038
Dream-I MBPP	0.257	0.186	0.176	0.076
MDLM-110M	0.002	0.009	0.0019	−0.0023

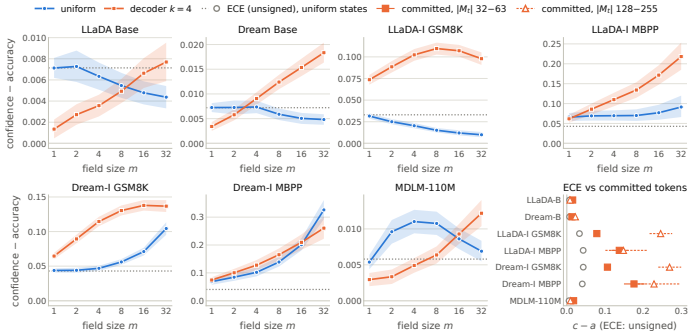
E1 positive: a token that won among 32 blanks is more overconfident than a random blank.

E2 positive: part of that growth is the winner's curse itself (Proposition 2).

> **Claims.** With a decoder choosing, E1 is positive for every model; E2 is clearly above the estimator's known bias (T0) only for the instruction-tuned models.

C1 results: with a decoder choosing, overconfidence grows with the contest

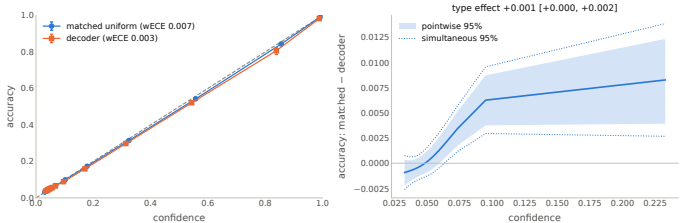
Gap Δ(*m*) against group size *m* for each model; right: the tokens a decoder commits



With a decoder choosing, E1 is positive and claimed for all models. For tokens committed while many blanks remain, overconfidence is multiple times the ECE measured on random blanks, across every model

C1 results: blanks a decoder leaves are right less often (E3)

Blanks a decoder leaves against random blanks with as many masks, at equal confidence



Right panel. Accuracy drop at equal confidence. LLaDA-8B-Base: the accuracy drop is statistically significant ($p < 0.001$). The decoder is fed the true tokens, confirming this drop stems from which blanks remain rather than decoder errors.

Positive in every model and step size ($p \leq 0.002$): small for base models, much larger for instruction-tuned models. A calibration curve fitted on random blanks systematically overstates accuracy during decoding.

C1 checks: do simpler explanations account for E1 and E2? (T8)

Each column adds one control; temperature means one confidence rescaling per mask rate

	E1 = Δ(32) − Δ(1)				E2 = field contrast		
LLaDA-8B-Base	plain	+ position	+ token type	+ temperature	plain	+ position	+ token type
uniform	−0.0028	−0.0027	0.0002‡	−0.0023‡	0.0009	0.0008‡	0.0032‡
decoder k=4	0.0063	0.0054‡	0.0076‡	0.0063‡	0.0030	−0.0032‡	0.0076‡

Reduced against full precision (test half of the documents): E1 differences across states are reported in the table.

Result. With a decoder choosing, E1 stays positive under every control across all models (largest change under temperature rescaling). LLaDA Base's small E1 on random blanks vanishes once token type is controlled, demonstrating the core effect's robustness. ‡: adding the control significantly changes the value; most do, by small amounts.

C2: a synthetic task where we know the true answer distribution

Our 110M model, fine-tuned on it, so (A) and (B) can be computed exactly at every step

- **Task**: fill in the values of JSON records built from real schemas (SchemaStore), multiple fields each, one token per value.
- **Link strength** ρ: fields come in pairs. With probability ρ, written at the start of the record, the second field (follower) is fixed by the first (leader); otherwise it is drawn independently. So we know p^* exactly.
- **Model**: our 110M model, fine-tuned and tested across multiple ρ settings.
- **Decoding**: all values start blank; each step a rule picks k fields, the most confident or at random, and we fill them with the true values.
- **Check**: **(A)** + **(B)** equals the directly computed total error within numerical tolerance.

one test record (ρ = 1), values blank

```
{"_rho":"1.00",
  "bundle":[M], "path":[M],
  "ref":[M], "repo":[M],
  "stats":[M], "version":[M]}
```

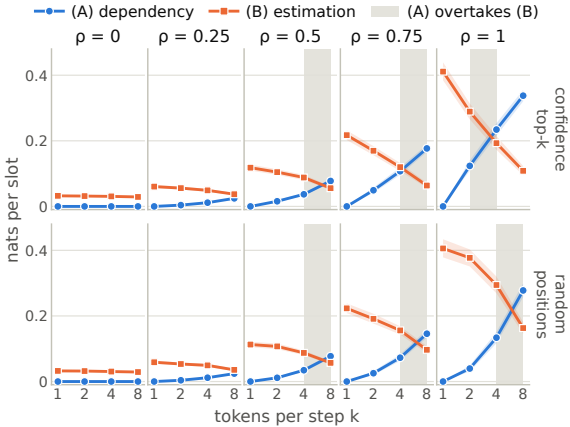
Per step, for the committed set S

(A) = $\sum_{i \in S} H(p_i^*) - H(p_S^*)$: zero unless a leader and its follower are committed together.

(B) = $\sum_{i \in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})$: each field's own error, including probability put on impossible values.

Summed over steps, per field, with bootstrap intervals over records.

C2: dependency errors (A) dominate only with strong links and large steps



- **(A)** is negligible at step size 1 or without link strength ($\rho = 0$), and grows with both.
- For strong links, **(A)** overtakes **(B)** as step size increases; picking by confidence causes this crossover at smaller step sizes.
- For weak links, estimation error **(B)** remains larger across step sizes.

Strong links ($\rho = 1$), large steps, confidence top-k

Dependency error **(A)** accounts for the vast majority of total error, far exceeding estimation error **(B)**.

C2: three findings that correct our plan

1 · Picking by confidence raises (A), not (B)

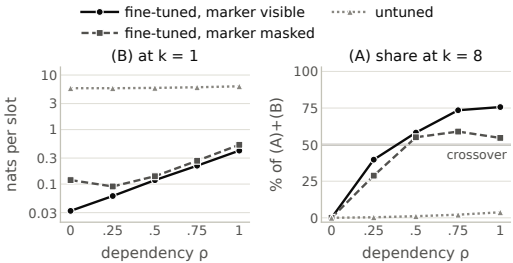
Expected: picking the surest fields would raise **(B)**.
Where **(B)** is compared, intervals overlap; **(A)** rises substantially compared to random selection.

2 · (B) also depends on ρ

Expected: ρ changes only **(A)**.
Even at step size 1, **(B)** increases with link strength ρ: the model only partly uses the ρ marker.

3 · (A) and (B) trade off as k grows

With strong links, as step size grows, **(B)** decreases while **(A)** increases: error from co-committed tokens shifts into **(A)**.



- Without fine-tuning, estimation error **(B)** dominates everywhere, with **(A)** remaining a minor fraction of the total.
- Picking by confidence fills a leader and its follower in the same step more often than chance, so **(A)** overtakes **(B)** at smaller k.

C2 on real text: little sign of dependency errors

On web-text documents (FineWeb-Edu); the proxy is checked on the synthetic task first

Proxy indicators on web text

For **(A)**: **disagreement**. Fill the k surest blanks all at once, then one at a time, and count how often a token changes. For **(B)**: the gap $\Delta = c - a$ and the cross-entropy (CE) of the true token.

Validation on synthetic task

Disagreement proxy correlates with step size k, reflecting error growth, while showing stability across link strengths ρ.

mask rates 0.1–0.9	110M (fp32)	LLaDA-8B Base (fp16)
disagreement, $k = 4$	0.0003	0.0001
disagreement, $k = 8$	0.0021	0.0022
CE (nats), $k = 1 \rightarrow 8$	0.08 → 0.40	0.08 → 0.23
Δ, $k = 1$	+0.001 [−0.003, 0.005]	+0.003 [−0.001, 0.006]
Δ, $k = 8$	+0.007 [0.004, 0.010]	+0.004 [0.002, 0.007]

CE also counts the text's own randomness, and blanks lower in the ranking are harder.

At standard decoding step sizes, real text consistently aligns with **weak link regimes, where estimation error (B) dominates**.

C3a: let corrected confidence decide how many tokens to commit

Uses only numbers the decoder already computes, so no extra model calls

- **Order**: rank the blanks of the current block by the model's own score ℓ, as top-k does.
- **Count**: commit the top n, where n is the number of blanks whose corrected confidence $\hat{c} \geq 1 - \alpha$ (at least one).
- **Corrector**: a small logistic model that also sees the median score, the share still masked and the number of blanks.
- **Fit once, then frozen**: on the fit halves of LLaDA-8B-Instruct runs on GSM8K and MBPP ($k = 2, 4, 8$ pooled); test halves never used.
- **Known shift**: real decoding starts from 256 blanks, while fit states sat inside a reference answer; we count steps outside the fit range.

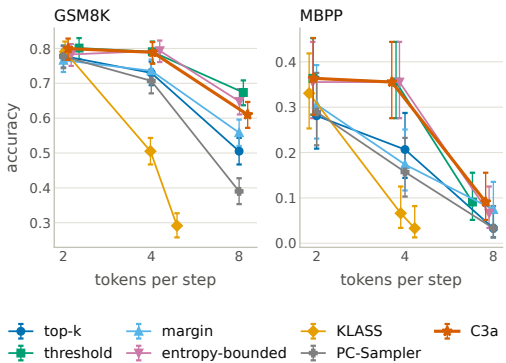
Corrector (T11)

	GSM8K	MBPP
fit documents	660	249
ECE, raw $\sigma(\ell)$	0.060	0.040
ECE, corrected $\hat{c}$	0.004	0.005
states where $\hat{c}$ rises with ℓ	1.000	1.000
decode steps outside fit range	0.017	0.034

ECE on held-out report-half states (teacher-forced), as in §6.

C3: is the corrected rule more accurate at the same speed?

LLaDA-8B-Instruct answers GSM8K and MBPP test problems itself



Decoders. Top-k, Fast-dLLM threshold, entropy-bounded, margin, KLASS, PC-Sampler, and C3a, each at three speeds.

Reading. Up and to the right is better. Speed is tokens per step, so compare accuracy at the same x.

Claim. A gain needs higher accuracy at the same speed, beyond the intervals.

Answer: no more accurate than the raw threshold rule; ahead of top-k, margin and PC-Sampler (as is the raw threshold).

What we got wrong, and the limits of these results

Claims we checked

- Unequal noise across blanks alone makes winners overconfident (initial premise): false, Proposition 1.
- Picking by confidence inflates **(B)**: on the synthetic task it raises **(A)** instead.
- ρ changes only **(A)**: **(B)** rises with ρ too.
- Real-text C2 as exact measurements: they are proxies that give a direction but no split in nats.

Limits

- C2's exact values hold for one synthetic task and one 110M model.
- Neither fine-tuned model passed our check that it reads the ρ marker; we use the one with lower validation loss (the other: backup).
- The disagreement proxy follows **(A)** weakly, and mostly through k.
- C1 is teacher-forced (the decoder sees the true tokens); E2 intervals treat the fitted calibration curves as fixed.
- C3: one model, greedy decoding, one seed

Summary

1 · Theory

Choosing a token cannot by itself make a calibrated model overconfident (Proposition 1). Any winner's curse is in the field term (Proposition 2), which we measure; in T0 our method detects bias that ECE misses.

2 · C1: are chosen tokens overconfident?

With a decoder choosing, overconfidence grows with the contest in every model; the winner's-curse component is clear in instruct models. Committed tokens show substantially higher overconfidence than standard ECE.

3 · C2: how large are (A) and (B)?

On the synthetic task, **(A)** overtakes **(B)** only with strong links ($\rho \geq 0.5$) and many fields per step; picking by confidence raises **(A)**, not **(B)**. On real text across typical decoding speeds, the **(A)** proxy is near zero, as when links are weak.

4 · C3: a corrected commit rule

At equal speed on GSM8K, the corrected rule C3a is no more accurate than Fast-dLLM's raw threshold; both beat top-k, margin and PC-Sampler. Calibration makes α a per-token error target but adds no accuracy.

Backup: proofs

W_i: the event that $i \in P$ wins, $i = \arg\max_{j \in P} \ell_j$; ties broken by independent randomness U

Proposition 1

Suppose, given s, each pair (ℓ_i, a_i) is independent of $\ell_{-i} = (\ell_j)_{j \in M, j \neq i}$. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$.

Given s, $(\ell_i, a_i) \perp \ell_{-i}$ and $(P, U) \perp (\ell_j, a_j)_{j \in M}$, so $(\ell_i, a_i) \perp Y \mid s$, and by weak union $a_i \perp Y \mid Z$, i.e. $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$.

W_i is determined by (Y, Z), so for every bounded measurable h

$$\mathbb{E}[a_i h(Z) 1_{W_i}] = \mathbb{E}[\pi(\ell_i, s) h(Z) 1_{W_i}].$$

Dividing by $\Pr(W_i) > 0$ gives $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s.

Proposition 2

Exactly one W_i holds. Applying the display with $h \equiv 1$, summing over i and averaging over s:

$$\mathbb{E}_m[a] = \sum_i \mathbb{E}_m[a_i 1_{W_i}] = \sum_i \mathbb{E}_m[\pi(\ell_i, s) 1_{W_i}] = \mathbb{E}_m[\pi(\ell, s)],$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; add and subtract $\pi(\ell)$.

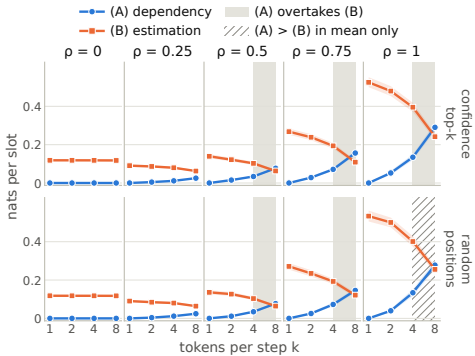
At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display, so the field term is 0.

Backup: how the main probe model was chosen

Recorded in `artifacts/probe/main_variant.json`

	ft110m-v3-visible	ft110m-v3
ρ marker in fine-tuning	never masked	masked
Fine-tuning tokens (M)	300	100
Marker effect on copy	0.251	0.014
Val. NELBO (nats/tok.)	0.1751	0.1892
First k with **(A)** > **(B)** (ρ=1)	4	8

- Neither model reaches the marker bar of 0.5, so the main model is the one with the lower validation NELBO.
- The other model crosses over at k = 8 for every $\rho \geq 0.5$ under both rules: the qualitative picture holds.



As the main figure, for ft110m-v3. Hatched: **(A)** > **(B)** in mean only.

Backup: disagreement by mask rate (T10)

Disagreement at $k = 8$ per mask rate; one state per document per rate

Mask rate	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
110M	0.0004	0	0	0	0	0	0	0.0013	0.0171	0.6250
LLaDA-8B-Base	0	0	0	0	0	0	0	0.0008	0.0192	0

- At rate 1 every input is the same all-`[MASK]` sequence. That one state gives the 110M model nearly all of its disagreement (0.0644 at $k = 8$ over all ten rates); LLaDA does not disagree there. The main slide reads rates 0.1–0.9.
- Documents under 256 tokens are skipped; at least 8 masked positions per state; 10,000 document-bootstrap resamples.

Backup: C1: the trace sweep

Collect once on GPU, one row per (masked state, position); draw pools and fit on CPU

Model	Text	Prec.	Docs	Rows (M)	GPU-h
LLaDA-8B Base	FineWeb-Edu	fp16	1,979	119.6	68
Dream-7B Base	FineWeb-Edu	fp16	1,976	119.4	60
Our 110M MDLM	FineWeb-Edu	fp32	1,979	119.6	3
LLaDA-8B Instruct	GSM8K	fp16	1,319	13.8	18
LLaDA-8B Instruct	MBPP	fp16	499	3.1	5
Dream-7B Instruct	GSM8K	fp16	1,319	13.4	14
Dream-7B Instruct	MBPP	fp16	499	2.5	4
LLaDA-8B Base	time-split	fp16	500	30.2	17
Dream-7B Base	time-split	fp16	500	30.2	15
Our 110M MDLM	time-split	fp32	500	30.2	1
LLaDA-8B Base	FineWeb-Edu	4-bit	198	12.0	5

Top: the primary arms. Grey: the unseen time-split set and the 4-bit control (T8). RTX 2080 Ti: 208 GPU-hours, 0.49 billion rows.

- **States per document**: uniform masks at rates 0.1–0.9; teacher-forced confidence top-k trajectories, $k \in \{2, 4, 8\}$; uniform masks matched to them (466 passes per document).
- **Per row**: candidate, confidence, logit, correctness, p(true), entropy, margin, top-20.
- **Halves**: 1,000 fit / 1,000 report documents. Calibrators, splines and knots from the fit half; every C1 number from the report half.
- **Deterministic**: repeats are bitwise identical, so the floor is 0 for every family.

Backup: data, contamination and our 110M model

- **2,000 calibration documents**, reservoir-sampled from FineWeb-Edu with a fixed seed and excluded from our 110M's training stream; 1,000 fit / 1,000 report.
- **Tasks**: GSM8K and MBPP test sets, split 50/50 the same way, as teacher-forced documents.
- **Contamination screen (T4)**: prompt with 50 tokens, continue greedily, flag a verbatim run ≥ 32 tokens. Flagged: 2/1,979 (LLaDA), 1/1,976 (Dream); the 8B arms are also analysed without them.
- **Time-split set**: text written after the 8B models were trained.

Model	bidirectional transformer: 12 layers, width 768, 12 heads; sequence length 256
Objective	MDLM (weighted masked cross-entropy), GPT-2 BPE + [MASK]
Training	169,055 steps, 10.0B tokens of FineWeb-Edu

37.2

perplexity *bound* on the 2,000 held-out docs [36.5, 38.0]

28.5

GPT-2 small, *exact* perplexity, same docs [28.0, 29.0]

MDLM gives only an upper bound, and GPT-2 saw other data with far more compute.

Backup: C1 primary endpoints, every arm, with intervals

95% intervals and the Holm decision in the 14-test families

Arm	Masks	E1	Holm	E2	Holm	β_z	β_z^{FE}
LLaDA Base	uniform	−0.003*,† [−0.004, −0.002]	yes	0.0010*,† [0.0009, 0.0010]	yes	−0.03‡	−0.01‡
	dec. k=4	0.006*,† [0.005, 0.008]	yes	0.0036*,† [0.0035, 0.0037]	yes	0.01‡	0.01
Dream Base	uniform	−0.002*,† [−0.004, −0.001]	yes	0.0011*,† [0.0011, 0.0012]	yes	−0.03‡	0.00
	dec. k=4	0.015*,† [0.013, 0.017]	yes	0.0029*,† [0.0028, 0.0029]	yes	−0.02‡	−0.02‡
LLaDA-I GSM8K	uniform	−0.022*,† [−0.025, −0.018]	yes	−0.003*,† [−0.004, −0.003]	yes	−0.16‡	−0.02
	dec. k=4	0.024*,† [0.016, 0.032]	yes	0.021*,† [0.020, 0.023]	yes	−0.10‡	−0.02
LLaDA-I MBPP	uniform	0.025† [0.003, 0.051]	no	0.013*,† [0.010, 0.016]	yes	−0.56‡	−0.23‡
	dec. k=4	0.156*,† [0.126, 0.189]	yes	0.036*,† [0.031, 0.041]	yes	−0.04‡	0.01
Dream-I GSM8K	uniform	0.061*,† [0.052, 0.070]	yes	0.021*,† [0.020, 0.023]	yes	−0.42‡	−0.03
	dec. k=4	0.072*,† [0.064, 0.081]	yes	0.038*,† [0.036, 0.039]	yes	−0.05‡	0.02‡
Dream-I MBPP	uniform	0.257*,† [0.227, 0.290]	yes	0.176*,† [0.159, 0.194]	yes	−0.76‡	−0.07
	dec. k=4	0.186*,† [0.151, 0.224]	yes	0.076*,† [0.066, 0.086]	yes	−0.19‡	0.01
MDLM-110M	uniform	0.00152† [0.00003, 0.00301]	no	0.0019*,† [0.0018, 0.0019]	yes	−0.01‡	0.00
	dec. k=4	0.009*,† [0.007, 0.011]	yes	−0.0023*,† [−0.0023, −0.0022]	yes	0.02‡	0.01

Backup: C1 diagnostic: does winning a larger field predict correctness? (β_z)

$\Pr(a_i{=}1) = \sigma(\alpha\,[+\alpha_s] + f(\ell_i) + \beta_z z_{r,m})$, without and with state effects α_s

	β_z, no state effects		β_z, state effects	
Arm	uniform	decoder	uniform	decoder
LLaDA Base	−0.030	0.014	−0.014	0.006
Dream Base	−0.027	−0.024	−0.002	−0.021
LLaDA-I GSM8K	−0.163	−0.095	−0.023	−0.020
LLaDA-I MBPP	−0.555	−0.042	−0.228	0.007
Dream-I GSM8K	−0.424	−0.053	−0.031	0.020
Dream-I MBPP	−0.764	−0.187	−0.065	0.007
MDLM-110M	−0.015	0.018	0.002	0.011

Field effect. $\beta_z < 0$ without state effects and ≈ 0 with them; it should agree with a positive E2.

Shared errors. $\beta_z \neq 0$ with state effects: positions of a state are wrong together, Proposition 1 fails, and E2 is not a field effect.

Caution. In T0, shared errors gave a *positive* β_z without state effects and a negative field term, so we read β_z only beside the sign of E2.

> **Reading.** No state effects: excludes 0 in 14 of 14, negative in 12. With them: holds 0 in 8 of those 12. A field effect, small where E2 is. Diagnostic only.

Backup: β_z with intervals

Without and with state effects; a diagnostic, no Holm family

Arm	Masks	β_z, no FE	β_z, FE
LLaDA Base	uniform	−0.03 [−0.04, −0.02]	−0.014 [−0.029, −0.001]
	dec. k=4	0.014 [0.004, 0.024]	0.01 [−0.01, 0.02]
Dream Base	uniform	−0.03 [−0.04, −0.01]	0.00 [−0.01, 0.01]
	dec. k=4	−0.02 [−0.03, −0.01]	−0.02 [−0.03, −0.01]
LLaDA-I GSM8K	uniform	−0.16 [−0.22, −0.11]	−0.02 [−0.08, 0.03]
	dec. k=4	−0.10 [−0.11, −0.08]	−0.02 [−0.04, 0.00]
LLaDA-I MBPP	uniform	−0.56 [−0.64, −0.47]	−0.23 [−0.33, −0.13]
	dec. k=4	−0.042 [−0.083, −0.002]	0.01 [−0.04, 0.05]
Dream-I GSM8K	uniform	−0.42 [−0.47, −0.38]	−0.03 [−0.07, 0.02]
	dec. k=4	−0.05 [−0.07, −0.03]	0.0204 [0.0001, 0.0413]
Dream-I MBPP	uniform	−0.76 [−0.86, −0.67]	−0.07 [−0.16, 0.03]
	dec. k=4	−0.19 [−0.24, −0.14]	0.01 [−0.05, 0.06]
MDLM-110M	uniform	−0.015 [−0.026, −0.003]	0.00 [−0.01, 0.01]
	dec. k=4	0.02 [0.01, 0.03]	0.01 [0.00, 0.02]
LLaDA Base TS	uniform	—	—
	dec. k=4	—	—
Dream Base TS	uniform	—	—
	dec. k=4	—	—
MDLM-110M TS	uniform	—	—
	dec. k=4	—	—

Backup: T8 controls, every arm

E1 and E2 without the control and with it; ‡: paired difference excludes 0

Arm	Masks	E1	+pos.	+type	+temp.	E2	+pos.	+type	+temp.
LLaDA Base	uniform	−0.003	−0.003‡	0.000‡	−0.002‡	0.001	0.001‡	0.003‡	—
	dec. k=4	0.006	0.005‡	0.008‡	0.006‡	0.003	−0.003‡	0.008‡	—
Dream Base	uniform	−0.002	−0.002	−0.001‡	−0.002‡	0.001	0.001‡	0.002‡	—
	dec. k=4	0.015	0.015‡	0.021‡	0.016‡	0.002	−0.001‡	0.008‡	—
LLaDA-I GSM8K	uniform	−0.022	−0.019‡	−0.022‡	−0.014‡	−0.003	−0.001‡	−0.004‡	—
	dec. k=4	0.024	0.022‡	0.023‡	0.028‡	0.021	0.015‡	0.022‡	—
LLaDA-I MBPP	uniform	0.025	0.022	0.024‡	0.014‡	0.013	0.037‡	0.012‡	—
	dec. k=4	0.156	0.153	0.153‡	0.125‡	0.039	0.069‡	0.033‡	—
Dream-I GSM8K	uniform	0.061	0.061	0.059‡	0.075‡	0.014	0.035‡	0.012‡	—
	dec. k=4	0.072	0.070‡	0.070‡	0.059‡	0.038	0.032‡	0.039‡	—
Dream-I MBPP	uniform	0.257	0.232‡	0.254	0.260	0.098	0.120‡	0.086‡	—
	dec. k=4	0.186	0.168‡	0.183‡	0.163‡	0.077	0.078	0.075‡	—
MDLM-110M	uniform	0.002	0.002‡	0.006‡	0.000‡	0.002	0.002‡	0.005‡	—
	dec. k=4	0.009	0.010‡	0.011‡	0.010‡	−0.003	−0.006‡	0.003‡	—

Backup: C1 robustness

None of these is in a Holm family

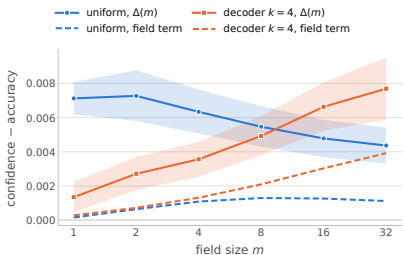
Arm	Masks	E1	E1 clean	E2, 5 kn.	E2, 9 kn.	E2, 13 kn.	E2 clean	E2 add.
LLaDA Base	uniform	−0.003†	−0.003†	0.001†	0.001†	0.001†	0.001†	0.001†
	dec. k=4	0.006†	0.006†	0.004†	0.004†	0.004†	0.004†	0.003†
Dream Base	uniform	−0.002†	−0.002†	0.001†	0.001†	0.001†	0.001†	0.001†
	dec. k=4	0.015†	0.015†	0.003†	0.003†	0.003†	0.003†	0.004†
LLaDA-I GSM8K	uniform	−0.022†	—	−0.003†	−0.003†	−0.003†	—	−0.002†
	dec. k=4	0.024†	—	0.021†	0.021†	0.021†	—	0.013†
LLaDA-I MBPP	uniform	0.025†	—	0.013†	0.013†	0.013†	—	0.000
	dec. k=4	0.156†	—	0.036†	0.036†	0.035†	—	0.007†
Dream-I GSM8K	uniform	0.061†	—	0.020†	0.021†	0.022†	—	0.005†
	dec. k=4	0.072†	—	0.038†	0.038†	0.037†	—	0.013†
Dream-I MBPP	uniform	0.257†	—	0.175†	0.176†	0.176†	—	0.098†
	dec. k=4	0.186†	—	0.076†	0.076†	0.075†	—	0.039†
MDLM-110M	uniform	0.002†	—	0.002†	0.002†	0.002†	—	0.002†
	dec. k=4	0.009†	—	−0.002†	−0.002†	−0.002†	—	0.001†
LLaDA Base TS	uniform	−0.001	—	—	0.002†	—	—	—
	dec. k=4	0.008†	—	—	−0.001†	—	—	—
Dream Base TS	uniform	0.002	—	—	0.004†	—	—	—
	dec. k=4	0.012†	—	—	0.001†	—	—	—
MDLM-110M TS	uniform	0.012†	—	—	0.005†	—	—	—
	dec. k=4	0.005†	—	—	0.000†	—	—	—

Masks	Effect	fp16	4-bit	Difference
uniform	E1	−0.002	−0.002	0.000 [−0.001, 0.002]
uniform	E2	−0.001	−0.001	0.0000 [−0.0001, 0.0001]
uniform	β_z, no FE	−0.070	−0.070	0.001 [−0.014, 0.016]
dec. k=4	E1	0.000	0.006	0.006 [0.000, 0.012]
dec. k=4	E2	0.008	0.006	−0.003‡ [−0.003, −0.002]
dec. k=4	β_z, no FE	0.002	0.000	−0.002 [−0.028, 0.024]

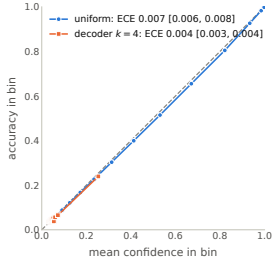
Reading. A column equal to the plain E1 or E2: the claim does not depend on the knots or on contaminated text. E2 uses a calibrator, E1 does not.

Backup: the main arm, its field term and reliability curve

Main arm: LLaDA-8B-Base; the other arms are in the backup



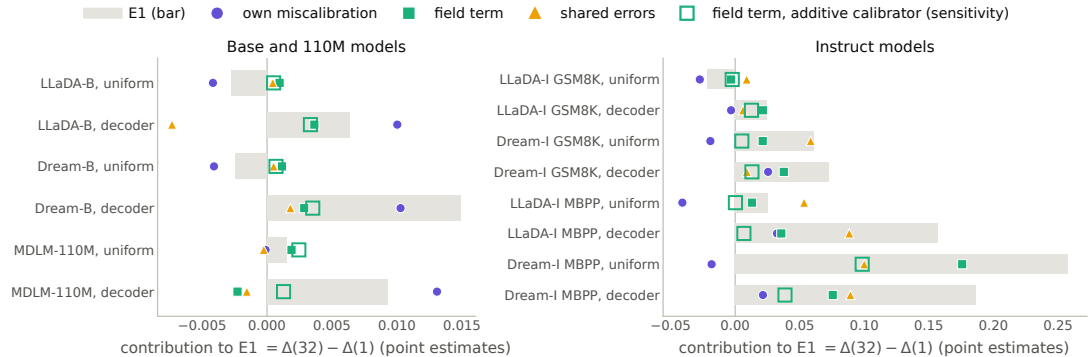
Left. Blue: uniform; orange: decoder states. Solid Δ(*m*), dashed its field term; the gap between them is own miscalibration. Rising solid: E1; rising dashed: E2.



Right. Below the diagonal: confidence exceeds accuracy. ECE sees only this picture, not the split into own and field terms (T0).

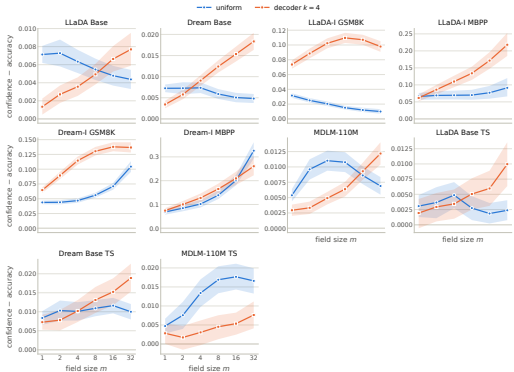
Backup: what makes E1 grow

E1 (bar) split into own miscalibration, field term and shared errors; the points sum to the bar



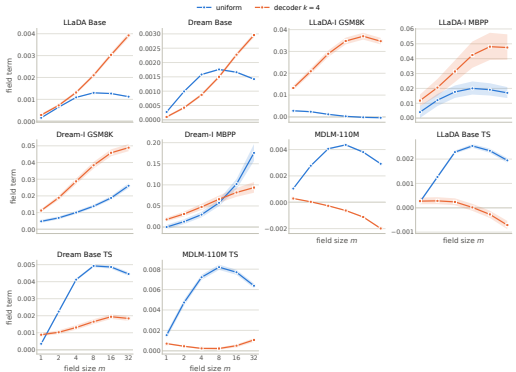
Backup: $\Delta(m)$ **in every arm**

Blue: uniform states; orange: decoder k=4; 95% bands



Backup: the field term in every arm

Blue: uniform states; orange: decoder k=4; 95% bands



Backup: E3 in every arm

Type effect at equal confidence against matched uniform states and the reweighted uniform grid

Arm	k	Matched uniform	p	Uniform grid	p
LLaDA Base	2	0.0010 [0.0002, 0.0019]	< 0.001	0.001 [0.000, 0.002]	0.006
	4	0.0011 [0.0003, 0.0020]	< 0.001	0.001 [0.000, 0.002]	< 0.001
	8	0.0014 [0.0004, 0.0024]	< 0.001	0.0013 [0.0002, 0.0023]	< 0.001
Dream Base	2	0.003 [0.002, 0.004]	< 0.001	0.003 [0.002, 0.005]	< 0.001
	4	0.003 [0.002, 0.004]	< 0.001	0.003 [0.002, 0.004]	< 0.001
	8	0.002 [0.001, 0.003]	< 0.001	0.003 [0.002, 0.004]	< 0.001
LLaDA-I GSM8K	2	0.056 [0.050, 0.062]	< 0.001	0.054 [0.047, 0.060]	< 0.001
	4	0.047 [0.041, 0.054]	< 0.001	0.046 [0.039, 0.053]	< 0.001
	8	0.030 [0.022, 0.038]	< 0.001	0.034 [0.027, 0.041]	< 0.001
LLaDA-I MBPP	2	0.046 [0.034, 0.059]	< 0.001	0.045 [0.032, 0.059]	< 0.001
	4	0.046 [0.030, 0.062]	< 0.001	0.040 [0.027, 0.053]	< 0.001
	8	0.031 [0.015, 0.047]	< 0.001	0.039 [0.025, 0.052]	< 0.001
Dream-I GSM8K	2	0.040 [0.034, 0.046]	< 0.001	0.041 [0.036, 0.047]	< 0.001
	4	0.043 [0.037, 0.049]	< 0.001	0.046 [0.040, 0.052]	< 0.001
	8	0.048 [0.042, 0.055]	< 0.001	0.050 [0.044, 0.056]	< 0.001
Dream-I MBPP	2	0.064 [0.049, 0.077]	< 0.001	0.066 [0.054, 0.078]	< 0.001
	4	0.063 [0.049, 0.077]	< 0.001	0.054 [0.041, 0.067]	< 0.001
	8	0.042 [0.025, 0.060]	0.002	0.034 [0.021, 0.048]	< 0.001
MDLM-110M	2	0.005 [0.004, 0.006]	< 0.001	0.004 [0.003, 0.005]	< 0.001
	4	0.006 [0.005, 0.007]	< 0.001	0.005 [0.004, 0.006]	< 0.001
	8	0.006 [0.005, 0.007]	< 0.001	0.005 [0.004, 0.006]	< 0.001

Backup: C3: does the harness reproduce known decoders, and at what cost?

Fewer steps need not mean less time: wall-clock and forward passes beside tokens per step

Sanity: GSM8K fit half, n = 200**, 5-shot, Fast-dLLM settings**

Top-k (k=1) ours 0.835 [0.777, 0.880] published 0.793

Threshold (τ=0.9) ours 0.845 [0.788, 0.889] published 0.792

Decoder	speed 1			speed 2			speed 3			problems		
	tok/st	NFE	s	tok/st	NFE	s	tok/st	NFE	s	err	t/o	uns
GSM8K (N = 659)												
Top-k	2.00	128.0	7.10	4.00	64.0	3.55	8.00	32.0	1.77	0	0	2
Threshold	2.26	113.2	6.27	4.03	63.5	3.52	8.27	30.9	1.72	0	0	1
Margin	2.00	128.0	7.10	4.00	64.0	3.55	8.00	32.0	1.77	0	0	6
Entropy	2.13	120.5	6.69	4.28	59.9	3.33	8.02	31.9	1.77	0	0	1
KLASS	2.02	126.4	10.76	3.97†	64.4	5.48	4.90†	52.2	4.42	0	0	9
PC-Sampler	2.00	128.0	7.09	4.00	64.0	3.55	8.00	32.0	1.77	0	0	3
C3a	2.08	123.4	9.12	4.04	63.3	4.69	8.56	29.9	2.21	0	0	0
C3a steps outside the fit range: 0.017												
MBPP (N = 121)												
Top-k	2.00	128.0	18.42	4.00	64.0	9.20	8.00	32.0	4.62	0	1	0
Threshold	1.94	131.8	19.19	3.73	68.7	10.04	6.79	37.7	5.50	0	0	0
Margin	2.00	128.0	18.46	4.00	64.0	9.19	8.00	32.0	4.59	0	1	0
Entropy	1.94	131.9	19.21	3.83	66.8	9.71	7.74	33.1	4.76	0	4	0
KLASS	1.89	135.7	26.54	3.87†	66.2	12.81	4.32†	59.3	11.42	0	0	0
PC-Sampler	2.00	128.0	18.33	4.00	64.0	9.19	8.00	32.0	4.61	0	1	0
C3a	1.95	131.4	22.68	3.61	71.0	12.52	7.53	34.0	5.96	0	0	0
C3a steps outside the fit range: 0.034												

err: driver errors (scored wrong; speed is over the problems that decoded); t/o: scoring timeouts (wrong); uns: no stop string in 256 tokens (not necessarily wrong); each summed over the three speeds. †: off the decoder's published operating point (deviation 17).

Reading. Both published accuracies lie inside our 95% intervals (top-k 0.835 against 0.793, threshold 0.845 against 0.792), though our points are higher. A decoder faster in steps but not in seconds is reported as such.

Backup: references

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